LEARNING THROUGH RECONSTRUCTION · DELIVERABLE 1 OF 8

Historical Timeline

From implicit model context to managed conversational memory, read as a chain of bottlenecks.

System Conversational Memory Intelligence System
Form approach → observed bottleneck → next approach

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HOW TO READ THIS

Each era below states what its approach **bought**, what limit it **hit**, and which limit **forced** the next move. The claim being made is not that this is the only possible reading of the literature. It is that every capability in the current design can be traced to a specific bottleneck someone actually ran into, rather than being copied from a reference architecture. Where a step is my interpretation rather than something a paper claims, it is tagged INFERENCE.

1. Knowledge in the weights, state in a hidden vector

BEFORE 2014 · RECURRENT SEQUENCE MODELS

Continuity across a sequence lives in a fixed-size hidden state carried from step to step, and everything else the model knows lives in its parameters. There is no addressable store and no write path other than gradient descent.

Bought: continuity within a sequence, learned end to end.
Hit: the hidden state is a fixed-size bottleneck that must compress unboundedly long history into constant space, and the only way to add a fact is to retrain.
→ **Forced:** memory that is external to the computation and addressable rather than blended into one vector.

2. Explicit external memory, differentiable and addressable

2014 TO 2015 · MEMORY NETWORKS; NEURAL TURING MACHINES; END-TO-END MEMORY NETWORKS

Separate the store from the controller. Give the model an array it can read from and write to via learned addressing, and let it take several hops before answering. VERIFIED This is the origin of the read/write/address decomposition that every later memory system inherits, including this one.

Bought: memory as a first-class, inspectable component, with multi-hop access.
Hit: the memory is fixed-size and trained jointly with the task, so it does not extend to millions of user-specific facts. Differentiable addressing over every slot is quadratic in store size. INFERENCE More decisively for this project, there is no lifecycle at all: no expiry, no supersession, no deletion, no provenance.
→ **Forced:** giving up differentiability of the store, and treating retrieval as a separate non-learned system component.

3. Attention replaces recurrence

2017 · ATTENTION IS ALL YOU NEED

Dispense with the recurrent state. Every position attends to every other position directly inside a fixed window.

Bought: full pairwise access within the window, and the scaling behaviour that made large pretrained models practical.
Hit: attention is quadratic in sequence length and the window is hard-bounded. INFERENCE Critically, the window is per-request. It has no persistence, no write path, and no existence after the response is returned. The transformer solved access *within* a context and left the question of what should be in that context entirely open.
→ **Forced:** deciding what to put in a finite window, which is a selection problem over an external corpus.

4. Retrieval augmentation: separate what is known from what is remembered

2020 · RETRIEVAL-AUGMENTED GENERATION

Keep a non-parametric index alongside the parametric model. Embed the query, retrieve the top-k passages, condition generation on them. VERIFIED The decisive contribution is that the knowledge source becomes editable without retraining.

Bought: a write path that does not require gradient descent, and provenance, because the retrieved passage can be shown.
Hit: the corpus is assumed static, internally consistent, and externally curated. Nothing writes to it from the conversation, ranking is a single similarity signal, and there is no notion of a fact ceasing to be true.
→ **Forced (two ways):** tighter model-level integration of retrieval, and separately, indexes that could actually hold billions of vectors.

5a. Retrieval pushed into the model

2022 · RETRO; MEMORIZING TRANSFORMERS

Retrieve during the forward pass rather than as a preprocessing step, either from a trillion-token database or from cached attention keys and values of earlier inputs.

Bought: evidence that a small model with a large retrieval store can match a much larger parametric model, which reframes memory as a substitute for parameters rather than an accessory to them.
Hit: still a read-only corpus, still no admission policy, and no user- or tenant-level semantics. INFERENCE These systems make retrieval better without making *memory* a managed object, which is the gap this project is actually in.
→ **Forced:** looking at ranking policy rather than retrieval mechanism.

5b. Indexes that make retrieval practical at scale

2017 TO 2018 · FAISS; HNSW

Approximate nearest-neighbour search over billions of vectors at interactive latency, via GPU-accelerated search and navigable small-world graphs.

Bought: the scale assumption everything above quietly depends on. Without this branch retrieval augmentation stays a research result.
Hit: these are index structures, not data-management systems. Deletion is awkward or deferred, incremental update is a rebuild-versus-recall trade-off, and attribute filtering fights recall. INFERENCE The absence of first-class deletion at the index layer is precisely why erasure has to be designed as a distributed operation with a stated consistency window rather than assumed to be a delete call.
→ **Forced:** treating the index as a rebuildable projection of a separate source of truth, not as the store itself.

6. Ranking becomes multi-signal, and memory gets a lifecycle

2023 · GENERATIVE AGENTS

A memory stream of observations, retrieved by a score combining relevance, recency, and importance, with periodic reflection that synthesises higher-level memories from lower-level ones. VERIFIED This is direct published evidence that single-signal similarity is insufficient, in a setting far less adversarial than a revenue system.

Bought: the two ideas this project depends on most. Ranking is a policy over several signals, and memory has a lifecycle rather than only a write and a read.
Hit: importance is scored by an LLM call at write time, which is expensive at production volume. INFERENCE More fundamentally, the setting is a simulation of agents with no real data subjects, so there is no PII admission problem, no tenant boundary, no erasure obligation, no supersession of a stated preference, and no offline evaluation set. Every one of those is load-bearing in my setting.
→ **Forced:** asking who manages the context budget, and separately, what happens when memory meets real users and real regulation.

7. Context as a managed resource

2023 · MEMGPT

Treat the context window as the analogue of main memory in an operating system, with an external store as disk, and let the model page information in and out under a budget.

Bought: the framing that context is an allocation problem with a budget and an allocator, rather than a container you fill.
Hit: the self-managed paging loop assumes the model is a trustworthy allocator, and it says nothing about which information was eligible to be stored in the first place. INFERENCE Admission, deletion guarantees, tenant isolation, and reproducible offline evaluation are all outside its scope. The useful transfer is the budget framing, not the paging mechanism.
→ **Forced:** an explicit admission stage before storage, and policy that lives outside the model rather than inside its loop.

8. The long-context detour, and why it does not close the loop

2023 ONWARD · LOST IN THE MIDDLE, AND THE GROWTH OF WINDOW SIZES

The obvious response to all of the above is to make the window large enough that selection stops mattering. VERIFIED Liu et al. show that recall from a long context is strongly position-dependent and worst when the relevant fact sits in the middle, so usable capacity is smaller than nominal capacity.

Bought: real headroom. Many problems that needed retrieval in 2020 do not need it now.
Hit: the quantitative limit is position-dependent recall. INFERENCE The categorical limit is more important: a window is per-request working memory. It has no persistence, no write path, no deletion semantics, no tenant boundary, and no audit trail, at any size. Growing it changes the size of the selection budget and nothing else about the structure of the problem.
→ **Forced:** accepting that memory is a system to be built, not a model capability to wait for.

9. The gap this project occupies

2026 · MANAGED CONVERSATIONAL MEMORY

Each era above solved the bottleneck in front of it and left a different one exposed. Read as a set, what is missing is not a mechanism but a *policy layer*: decisions about what is admitted, how it is represented, how long it stays true, who may see it, and how any of that is measured.

Inherited: read/write/address decomposition (era 2), an editable non-parametric store (era 4), practical ANN indexing (era 5b), multi-signal ranking and lifecycle (era 6), context as a budget (era 7).
Still open: admission control over ineligible and low-value information; typed facts with provenance, confidence, and validity intervals; conflict resolution over superseded claims; deletion with a tested consistency window; tenant isolation as an invariant rather than a filter; and reproducible offline evaluation against a baseline.
→ **This project.** The ten capabilities derived in `first_principles.md`.

Compressed chain

ERA	APPROACH	OBSERVED BOTTLENECK	WHAT IT FORCED
<2014	State in a hidden vector, knowledge in weights	Fixed-size state; the only write path is retraining	External addressable memory
2014–15	Differentiable external memory	Fixed-size, jointly trained, no lifecycle	Non-differentiable retrieval as a system component
2017	Attention over a fixed window	Quadratic cost; window is per-request and does not persist	Selection over an external corpus
2020	Retrieval-augmented generation	Static read-only corpus; single similarity signal	Better integration, and indexes at scale
2022	Retrieval inside the model	Still read-only; no user or tenant semantics	Attention shifts to ranking policy
2017–18	FAISS, HNSW	Deletion, incremental update, and filtering are index-level problems	Index as a rebuildable projection of a source of truth
2023	Generative Agents	Costly write-time importance; no privacy, tenancy, or evaluation	Multi-signal ranking, and a lifecycle
2023	MemGPT	No admission policy; model as its own allocator	Policy outside the model
2023+	Long context	Position-dependent recall; a window is not storage	Build the system
2026	**Managed conversational memory**	**To be established by the Deliverable 3 baseline** TO MEASURE	**C1 to C10**

HONEST CAVEAT

This chain is drawn to explain the design I am proposing, which means it is vulnerable to hindsight bias. The eras did not follow one another as cleanly as the table implies, and several were concurrent rather than sequential. The falsifiable version of the claim is narrower and is the one I will defend: **each of C1 to C10 addresses a bottleneck that at least one of these lines of work explicitly ran into or explicitly declared out of scope.** Anything in the design that cannot be traced to such a bottleneck should be cut in Deliverable 4.

Sources

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5. Johnson, J., Douze, M., Jégou, H. *Billion-scale Similarity Search with GPUs.* 2017. arXiv:1702.08734
6. Malkov, Y. A., Yashunin, D. A. *Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs.* 2018. arXiv:1603.09320
7. Lewis, P., Perez, E., Piktus, A., et al. *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.* 2020. arXiv:2005.11401
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9. Wu, Y., Rabe, M., Hutchins, D., Szegedy, C. *Memorizing Transformers.* 2022. arXiv:2203.08913
10. Park, J. S., O'Brien, J. C., Cai, C. J., et al. *Generative Agents: Interactive Simulacra of Human Behavior.* 2023. arXiv:2304.03442
11. Packer, C., Wooders, S., Lin, K., et al. *MemGPT: Towards LLMs as Operating Systems.* 2023. arXiv:2310.08560

12. Liu, N. F., Lin, K., Hewitt, J., et al. *Lost in the Middle: How Language Models Use Long Contexts.* 2023. arXiv:2307.03172

Items 1 to 11 are handbook Appendix G. Item 12 is added because it supplies the direct evidence for era 8.

AI assistance disclosure. Claude was used to draft prose and check the chain for gaps. The selection of eras, the bottleneck attributions, the separation of what each source claims from what I infer, and the caveat above are mine.